

K-CHOPORE — Results Report

A reproducible nanopore Direct RNA-seq pipeline for the plant m6A epitranscriptome, applied to the ANAC017-dependent mitochondrial retrograde response in *Arabidopsis thaliana*

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Abstract N6-methyladenosine (m6A) is the most abundant internal mRNA modification. Oxford Nanopore Direct RNA-seq (DRS) detects it on native transcripts, but reproducible, plant-ready workflows are scarce. We present **K-CHOPORE**, an end-to-end, containerised (Docker/FAIR) Snakemake pipeline from raw signal to m6A maps, differential expression and function. As a case study we profile the mitochondrial retrograde response of *Arabidopsis* (WT vs *anac017-1* × Control vs Antimycin A, 2×2, n=3). The pipeline recovers the canonical RRACH motif (validated against m6A-writer mutants), separates the design by expression, and reveals an ANAC017-dependent remodelling of the m6A epitranscriptome under mitochondrial stress.

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1 Introduction

m6A shapes transcript stability, splicing and translation. Direct RNA-seq reads modifications on native RNA without amplification, preserving the signal that error-based and signal-based callers exploit. Plant epitranscriptomics, however, lacks reproducible, containerised pipelines.

K-CHOPORE addresses this: one Snakemake workflow, one config, a Docker image with the full stack, and reuse of precomputed steps so only what changed re-runs. We demonstrate

it on **mitochondrial retrograde signalling**: Antimycin A inhibits complex III, triggering a retrograde response whose master regulator is the transcription factor **ANAC017**.

2 Methods

Design (2×2). Genotype (WT vs *anac017-1*) × treatment (Control vs Antimycin A), three biological replicates – 12 DRS libraries.

Pipeline. Basecalling (Guppy) → mapping to a FLAIR transcriptome (minimap2, -ax map-ont) → QC (NanoPlot, NanoComp, pycoQC, MultiQC) → isoforms (FLAIR, StringTie) → m6A by two orthogonal methods (ELIGOS2, error-based and genetically validated against m6A-writer mutants *mta/mtb/fip37/vir/hakai*; m6anet, signal-level) → differential expression (DESeq2, transcript-level, ~genotype+treatment+genotype:treatment) → functional enrichment (g:Profiler). Cross-condition comparisons use coverage-normalised rates. Per-tool settings and filters are listed in METHODS_TOOLS.md.

3 Results

3.1 Quality control

Read quality and yield are consistent across the 12 libraries (NanoComp, Fig. 1).

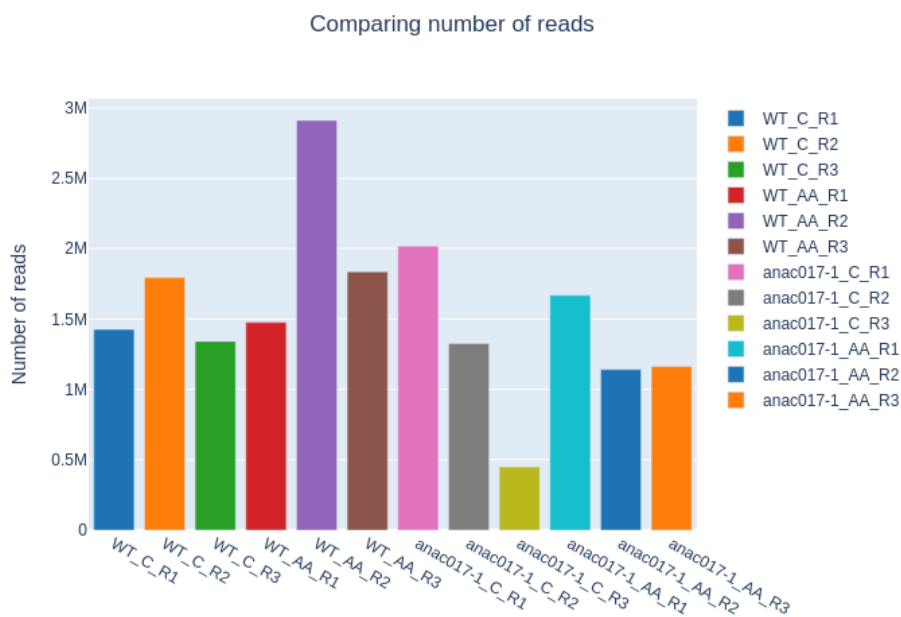


Figure 1: NanoComp comparison across the 12 DRS libraries.

3.2 Expression captures the design

PCA on variance-stabilised counts separates genotype (PC1) and treatment (PC2), with replicates clustering tightly (Figure 2). Differential expression yields 175 transcripts responding to Antimycin A, 127 with a basal genotype effect, and **32 transcripts with an ANAC017-dependent response** (genotype×treatment interaction; Figure 3).

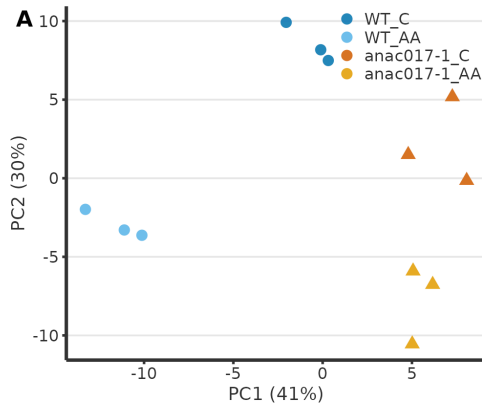


Figure 2: PCA, 2x2 design.

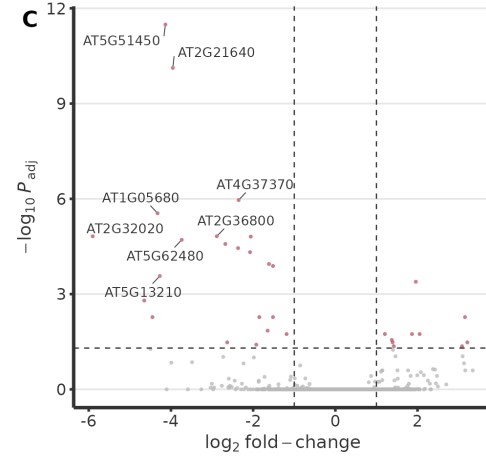


Figure 3: Interaction volcano (ANAC017-dependent).

3.3 The m6A epitranscriptome

ELIGOS2 recovers the canonical **RRACH** motif in all conditions, re-centred on the modified adenosine (Figure 4). After coverage normalisation, m6A density is comparable across conditions (276–352 sites per 1,000 tested adenosines), with *anac017-1* showing the highest basal rate (Figure 5) – the raw counts alone would have been misleading due to depth differences.

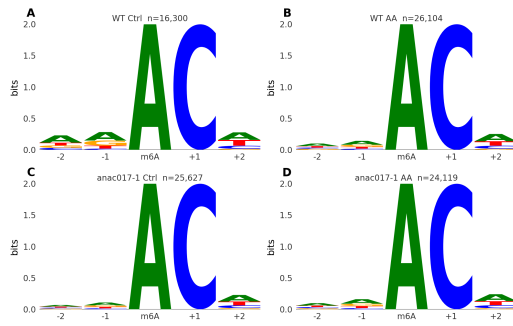


Figure 4: m6A motif (RRACH) by condition.

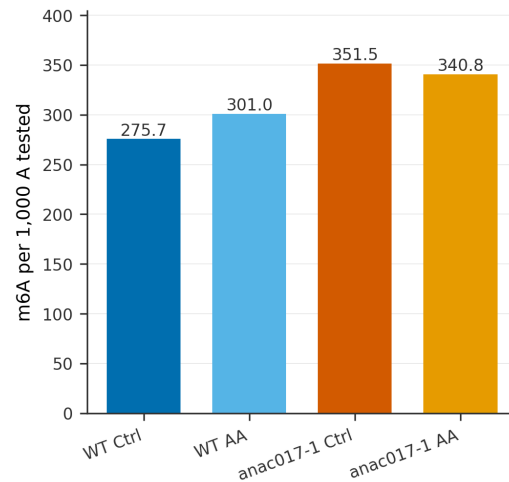


Figure 5: Coverage-normalised m6A rate.

At the retrograde marker **AOX1a** (AT3G22370), m6A signal increases at the 3' end under Antimycin A (Figure 6). Across the genome, ~89% of AA-gained m6A sites in WT are not gained in *anac017-1*, i.e. **ANAC017-dependent**.

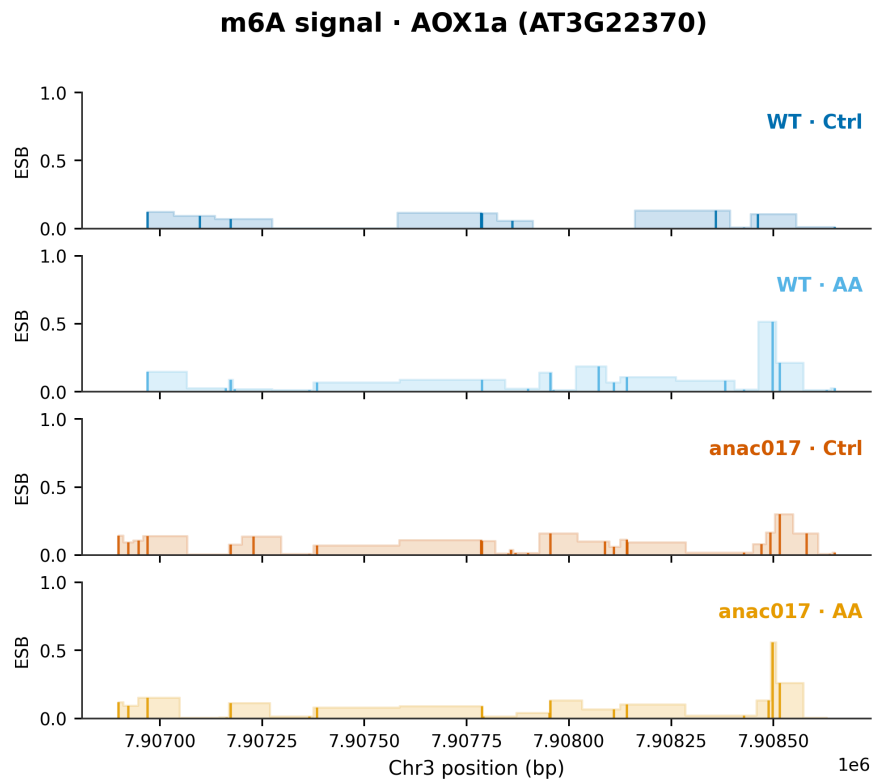


Figure 6: m6A signal along AOX1a in the four conditions.

3.4 Function

ANAC017-dependent genes are enriched in response to stimulus / stress and hypoxia (e.g. *CYP81D8*), consistent with complex-III inhibition mimicking an energy/oxygen stress (Figure 7).

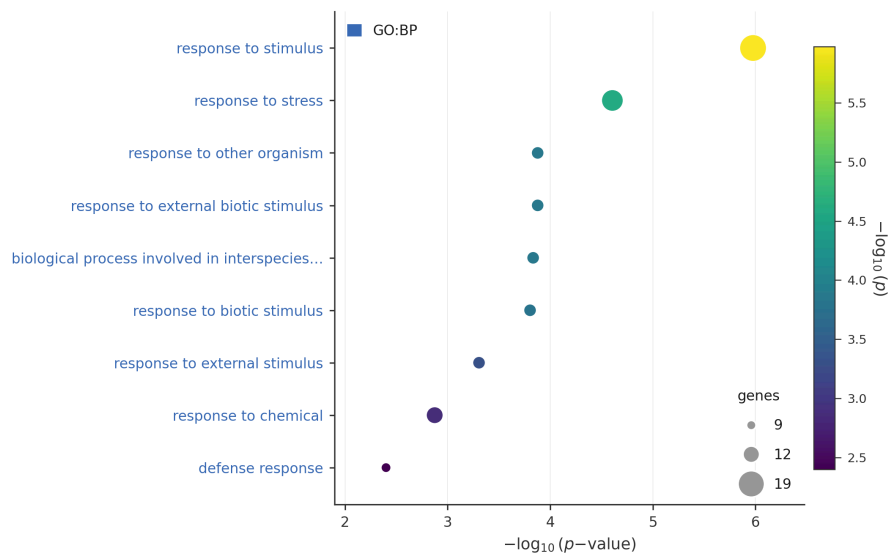


Figure 7: GO enrichment of ANAC017-dependent genes.

4 Discussion

The results sketch a model in which **mitochondrial stress reshapes the plant m6A epitranscriptome through ANAC017-dependent retrograde signalling**. The interaction term isolates the transcripts whose AA response requires ANAC017, and AOX1a — the canonical retrograde target — gains m6A under stress. Coverage normalisation was essential: comparing raw site counts would have confounded biology with sequencing depth. This is a follow-up study; the value here is the reproducible route from signal to a coverage-controlled m6A map.

5 Reproducibility (FAIR)

- **Code:** github.com/biopelayo/kchopore-anac017-drs
- **Container:** Docker image with the full stack (Snakemake, minimap2, samtools, FLAIR, ELIGOS2, m6anet, DESeq2, g:Profiler...).
- **Run:** `clone → docker run ... snakemake --configfile config/config_transcriptome.yml`. Editing only the config defines a new run; precomputed steps are reused.

6 Data & code availability

Pipeline, scripts, Docker recipe and this report: github.com/biopelayo/kchopore-anac017-drs. Raw DRS signal is large and shared on request / public repository (ENA/GEO).